APSMO 2010 Set 3

1 Problems

Problem 1.1 Joshua writes a four-digit number whose digits are $3, 5, 7$ and 9, not necessarily in that order. The number is a multiple of 5. The first two digits and the last two digits have the same sum. The thousands digit is larger than the hundreds digit.

What is Joshua's number?

Problem 1.2 One hat and two shirts cost \$21. Two hats and one shirt cost \$18. Megan has exatly enough money to buy one hat and one shirt.

How much money does Megan have?

Problem 1.3 It takes 3 painters 4 hours to paint 1 classroom. How many hours does it take 1 painter to paint 2 classrooms of the same size as the first one?

Problem 1.4 Mr. Wright wants to tile a 5m by 5m square floor. He has three kinds of square tiles: 1m by 1m, 2m by 2m, and 3m by 3m. Tiles may not overlap or be cut.

What is the fewest tiles Mr. Wright may use to completely cover his floor?

Problem 1.5 111111 is the product of 5 different prime numbers.

What is the sum of those 5 prime numbers?